ECE2111 Cheat Sheet

Signals and Systems. Methods, conditions, traps and past-paper answers. Tables are in the exam. From your weekly notes, the 2022 and 2024 papers and solutions, the sample and mock papers, the mid-sem and the formula sheet. Worked numbers rechecked in Python.

1 How to score, and what is handed to you

Read once before you start.

Section B: formula → substitution → answer → one-line reason
Trap: A bare number gets about 1 of 4.

Given: sifting, convolution, $\Sigma\alpha^k$, $\int e^x$, $\int x^k$, $X_s(\omega)$ formula, and all FS / FT / Laplace / Z tables
Trap: Use this sheet for methods, conditions and traps.

NOT in the tables: $-e^{-at}u(-t) \Leftrightarrow 1/(s+a)$, Re(s) < −Re(a). $-\alpha^n u[-n-1] \Leftrightarrow 1/(1-\alpha z^{-1})$, $|z| < |\alpha|$
Needs: Left-sided signals.

Table typo: DT FS square wave (|n| ≤ N, period N_0) is $X_k = \sin(k\omega_0(N+\tfrac{1}{2}))\,/\,(N_0 \sin(k\omega_0/2))$
Needs: Checked numerically.
Trap: Printed sheet drops $1/N_0$ and the sin in the denominator.

CT FS square wave from table: $X_k = \sin(k\omega_0 T)/(k\pi)$, $X_0 = 2T/T_0$
Needs: Pulse |t| < T in period T_0.
Trap: X_0 is the k→0 limit, not on the sheet.

Quantity	Hz	rad/s	rad/sample
Signal frequency	f	$\omega = 2\pi f$	$\omega T_s = 2\pi f/f_s$
Sampling rate	$f_s = 1/T_s$	$\omega_s = 2\pi/T_s$	2π
Nyquist frequency	$f_s/2$	π/T_s	π
Nyquist rate (bandwidth W)	2f_max	2W	—

2 Which tool? Read the question, pick the section

"Linear / TI / causal / memoryless / stable / real?" → 3. "Find T_0 or N_0" → 2

"Find y given x and h", "step response", deltas in h → 4

Periodic input, X_k, duty cycle, harmonics through a filter → 5

Sinusoid in/out, H(ω) from an equation, DC input → 6. X(ω) as a fraction, DTFT of $n\alpha^n$ → 7

Sampling rate, Nyquist, anti-aliasing, sketch X_s, ZOH → 8

Block diagram, H(s), RoC, BIBO, output (CT) → 9. Difference eq., H(z), polar poles, IIR/FIR (DT) → 10

Standard system, filter type, pole behaviour, partial fractions, any property in any domain → 11

3 Signals: transformations, periodicity, symmetry

2022 Q1, 2024 Q2, mock Q6

x(t−τ): τ>0 shifts RIGHT. x(αt): |α|>1 compresses
Trap: "+" inside the bracket moves left. Check one point.

x(−t+2) = x(−(t−2)): reflect first, then shift right 2
Trap: Check y(0) = x(2).

CT $e^{j\omega_0 t}$: $T_0 = 2\pi/|\omega_0|$, always periodic

DT $e^{j\omega_0 n}$: periodic iff $\omega_0/2\pi = a/b$ (rational, lowest terms) ⇒ $N_0 = b$
Needs: N must be an integer.
Trap: sin(4πn/5): $\omega_0/2\pi = 2/5 \Rightarrow N_0 = 5$, fundamental frequency 2π/5 (not 4π/5). cos(√2 n/2) is aperiodic.

Sum of sinusoids: f_0 = gcd of the frequencies; T_0 = lcm of the periods
Needs: CT ratio must be rational.
Trap: DT sums are always periodic.

Even x(−t)=x(t), odd x(−t)=−x(t). $x_e = [x(t)+x(-t)]/2$, $x_o = [x(t)-x(-t)]/2$
Trap: odd·odd = even, even·odd = odd.

Bounded: |x| ≤ M for all t (use |x| for complex)
Trap: $|e^{j(\pi/2)n + \cos n}| = e^{\cos n} \le e$: bounded.

Periodic extension of a finite x: T_0 can be SHORTER than the window
Needs: Look for repetition inside x.
Trap: DT window length is B−A+1; CT is b−a.

Past paper: 2022 Q1 at 10 samples/s, n=0..5: cos 2πt → 1, 0.81, 0.31, −0.31, −0.81, −1. sin 10πt → all 0. u(t−0.25) → 0,0,0,1,1,1. sin(2πt)u(t) → 0, 0.59, 0.95, 0.95, 0.59, 0.

4 System properties (test each separately)

every paper

Property	From the equation	From h (LTI)	From H(s) / H(z)
Linear	zero in → zero out; no x^2, |x|, +const, e^x	—	—
Time-inv.	no t or n multiplying x; no x(2t); no fixed-time condition	—	—
Causal	no x(t+·), x[n+·] (future)	h = 0 for t < 0	RoC right of rightmost pole / |z| > outermost pole
Stable	bounded in ⇒ bounded out; one counterexample disproves	$\int\|h\| < \infty$, $\Sigma\|h\| < \infty$	RoC contains jω axis / unit circle
Memoryless	only x at the same instant	h = αδ	H constant
Real	real coefficients, no $e^{j\omega t}$ multipliers	h real	poles and zeros in conjugate pairs
Causal+stable	—	both above	all poles Re(s) < 0 / |z| < 1

x times a fixed function of t (e^{-t}, sin t, n^2): linear but TV. LTI never creates new frequencies
Trap: Integrator and differentiator are unstable. 1/t is not integrable. Stable ≠ causal.

Series $h_1 * h_2$ (H_1H_2). Parallel h_1+h_2. Feedback $H_1/(1-H_1H_2)$
Trap: Negative feedback: H_2 includes the minus sign ⇒ $H_1/(1+H_1G)$.

Difference/differential eq. + "initial rest" ⇒ causal LTI
Trap: Without it the system is not fixed.

Past paper: 2024 Q1 $y=2e^{j\omega_1 t}x(t-1)+e^{j\omega_2 t}x(t-2)$: linear Y, TI N, causal Y, memoryless N, stable Y, real N.

Past paper: Mid-sem: $[x[n-1]]^2+x[n]$ → not linear, memory, TI, causal. $e^{-t}x(t+2)$ → linear, memory, TV, non-causal. 2022: $\Sigma_{k=-1}^{1} x[n-k]$ → linear, TI, memory, stable, non-causal.

5 Convolution and impulse response

mid-sem, 2024 Q2

$y[n] = \Sigma\, x[k]h[n-k]$. $y(t) = \int x(\tau)h(t-\tau)d\tau$
Needs: LTI.
Trap: Flip h, then slide. Split into overlap cases.

Support of $x*h = [a_1+a_2, b_1+b_2]$
Needs: Finite lengths: L_1+L_2-1 samples.

FIR equation $y = \Sigma b_k x[n-k] \Rightarrow h = \Sigma b_k \delta[n-k]$
Trap: Mock Q1: h = δ[n] − 0.5δ[n−1] − 0.5δ[n−2] + δ[n−3].

$x*\delta(t-t_0) = x(t-t_0)$: each delta in h gives a delayed copy of x
Trap: h = δ(t−1)−δ(t−2)+δ(t−3) ⇒ y = x(t−1)−x(t−2)+x(t−3). Stable (finite impulses).

Step response s = h∗u; h = ds/dt. y with u input: s(t) = u(t−1)−u(t−2)+u(t−3) for the h above

Past paper: 2024 Q2: z=u(t+1)−u(t−1), T_s=2/3 ⇒ x[−2..1] = 0,1,1,1 (x=1 for n=−1,0,1). $y[n] = \Sigma_{k=-1}^{1} h[n-k]$ ⇒ A=−1, B=1. Q2c diagram (final gain 0.1): a=1, b=2, c=10, d=0.5.

6 Fourier series questions

2022 Q3, 2024 Q4, mock Q3–4, sample Q8

$X_0 = (1/T_0)\int$ one period of x = average value
Trap: Duty cycle β, height 1 ⇒ $X_0 = \beta$.

Periodic x through LTI: $Y_k = H(k\omega_0)X_k$. DC: $Y_0 = H(0)X_0$
Needs: Stable LTI.
Trap: $\omega_0 = 2\pi/T_0$, not 2π/T of some other signal.

$H(0) = \int h(t)dt$ (= DC gain)
Trap: $h = e^{-2t}u - e^{-4t}u \Rightarrow H(0) = 1/2 - 1/4 = 1/4$.

Which harmonics pass a filter: list $k\omega_0$ for k = ±1, ±2, … and compare with the passband
Needs: Include negative k.
Trap: Two or more pairs pass ⇒ output is not a single cosine.

Real x, pair ±k passes: $y = 2|X_k| \cos(k\omega_0 t + \angle X_k)$

Parseval: $(1/T_0)\int|x|^2 = \Sigma|X_k|^2$. Real x: $\Sigma_{k\ge1}|X_k|^2 = (P - |X_0|^2)/2$
Trap: Mid-sem: P = 4, X_0 = 0 ⇒ 2.

Modulation by $\cos(M\omega_0 t)$: $W_k = \tfrac{1}{2}X_{k-M} + \tfrac{1}{2}X_{k+M}$
Trap: Demodulator: linear, NOT time-invariant.

Read X_k off Euler: $\cos\theta = \tfrac{1}{2}e^{j\theta} + \tfrac{1}{2}e^{-j\theta}$
Trap: No integral needed if x is a sum of harmonics.

$H(k\omega_0) = 0$ at some k ⇒ input not unique from output
Trap: Add a cosine at that frequency to x (2024 Q4d).

Differential eq. from H(ω): cross-multiply, jω → d/dt, $(j\omega)^2 = -\omega^2 \to d^2/dt^2$
Needs: State "initial rest".
Trap: 2024 Q4a: y″ + a y′ + y = x + x″/(4π²).

Past paper: 2022 Q3: x = 1−t² on |t|≤1, T_0=2 ⇒ X_0 = 2/3; with H(0)=1/4 ⇒ Y_0 = 1/6. 2024 Q4: X_0=β, H(0)=1, Y_0=1/4 ⇒ β=1/4.

Past paper: Sample Q8: bandpass passes only ω = ±1. T=2π ⇒ k=±1 ⇒ y = (2/π)sin(π/5)cos t. T=4π ⇒ k=±2 ⇒ y = (1/π)sin(2π/5)cos t. T=8π ⇒ several k ⇒ not one cosine.

Past paper: Mock Q4: $W_k = \tfrac{1}{2}X_{k-1}+\tfrac{1}{2}X_{k+1}$, $Y_0 = A\cdot W_0 = A(0.25+0.25) = 2 \Rightarrow A = 4$, any 0 < B < ω_c.

7 Frequency response

2022 Q4, sample Q6, mock Q6

$x = e^{j\omega t} \Rightarrow y = H(\omega)e^{j\omega t}$. $x = \cos(\omega t+\varphi) \Rightarrow y = |H(\omega)|\cos(\omega t+\varphi+\angle H(\omega))$
Needs: Real, stable LTI. Eternal input.
Trap: Frequency never changes. y = H(ω)x is wrong for general x.

Constant input a ⇒ y = a·H(0)
Needs: a = a·e^{j0t}.

Output sinusoid given, find input: divide each phasor by $H(\omega_0)$
Trap: Do each frequency separately.

From ODE: $\Sigma a_k(j\omega)^k Y = \Sigma b_k(j\omega)^k X$. From difference eq: $\Sigma a_k e^{-j\omega k} Y = \Sigma b_k e^{-j\omega k} X$
Trap: RC: H = 1/(1+jωRC).

Delay τ: $e^{-j\omega\tau}$. Gain 0.5 + delay 0.1: $h = 0.5\delta(t-0.1)$, $H = 0.5e^{-0.1j\omega}$
Trap: Undo it: z(t) = 2y(t+0.1).

Ideal LPF |ω|<W: h = sin(Wt)/(πt)
Trap: Non-causal. sin(Wt)/(πt) has cutoff W rad/s.

H = Y/X fails where X = 0
Trap: That is 0/0, not H = 0.

Past paper: Sample Q6: H = Y/X = $e^{j\omega}$(jω+1)/(jω+a); input a ⇒ y = a·H(0) = 1.

8 Fourier transform algebra (tables are given)

2022 Q5, 2024 Q7, mock Q3, sample Q4

Signal	Tool	Spectrum
CT periodic	CT Fourier series X_k	discrete, not periodic
DT periodic	DT Fourier series X_k	discrete, periodic in k (N_0)
CT aperiodic	CTFT X(ω)	continuous, not periodic
DT aperiodic	DTFT X(ω)	continuous, 2π-periodic
Not abs. integrable / summable	Laplace / Z (with RoC)	—

Equal-degree fraction: divide first. (4+jω)/(5+jω) = 1 − 1/(5+jω) ⇒ $\delta(t) - e^{-5t}u(t)$
Trap: The constant term is an impulse.

$n\alpha^n u[n] = (n+1)\alpha^n u[n] - \alpha^n u[n] \Rightarrow 1/(1-\alpha e^{-j\omega})^2 - 1/(1-\alpha e^{-j\omega})$
Needs: |α|<1. Checked numerically.

$\alpha^n \cos(n)u[n] \Rightarrow \tfrac{1}{2}/(1-\alpha e^{-j(\omega-1)}) + \tfrac{1}{2}/(1-\alpha e^{-j(\omega+1)})$
Needs: Euler, then modulation.

$\cos(\omega_0 t)x(t) \Rightarrow \tfrac{1}{2}X(\omega-\omega_0) + \tfrac{1}{2}X(\omega+\omega_0)$
Trap: Mock Q3: 10 periods in length 2 ⇒ period 0.2 ⇒ a = 10.

Periodic signal: $X(\omega) = 2\pi\Sigma X_k \delta(\omega - k\omega_0)$
Trap: No ordinary FT exists.

Parseval: $\int|x|^2dt = (1/2\pi)\int|X|^2d\omega$. $\int 1/(1+\omega^2)d\omega = 2\pi\int e^{-2t}dt = \pi$
Trap: Use it to avoid a hard integral.

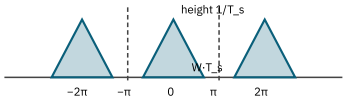
$H(\omega) = \Sigma\, h[n]e^{-j\omega n}$: rewrite cosines with Euler, coefficient of $e^{-j\omega n}$ is h[n]
Trap: 1.5+0.5cos ω+cos 2ω ⇒ h[0]=1.5, h[±1]=0.25, h[±2]=0.5. Non-causal. Times $e^{-j2\omega}$ ⇒ causal.

DT window u[n+N]−u[n−N] (2N samples) ⇔ $[\sin(N\omega)/\sin(\omega/2)]e^{j\omega/2}$
Needs: Current formula sheet.
Trap: 2024 paper printed sin(ω(N+½))/sin(ω/2): that is the 2N+1-sample window.

Past paper: 2024 Q7: Y=(4+jω)/(5+jω) ⇒ y=δ−e^{-5t}u. X=Y/H= (2+jω)/(5+jω)=1−3/(5+jω) ⇒ x=δ(t)−3e^{-5t}u(t).

9 Sampling, aliasing, reconstruction

2022 Q6, 2024 Q6, mock Q8, sample Q1, Q10



X copied every 2π, edge at W-T_s. No aliasing iff W-T_s < π.

No aliasing iff $T_s < \pi/W$ ($\omega_s > 2W$, $f_s > 2f_max$)
Needs: W = highest frequency in rad/s.
Trap: Bandpass 5π–8π: W = 8π ⇒ Nyquist rate 16π rad/s = 8 samples/s.

Nyquist rate = 2W (a rate). Nyquist frequency = π/T_s
Trap: Never sample a sinusoid at exactly $T_s = \pi/\omega_o$.

Anti-aliasing LPF cutoff ≤ π f_s rad/s (= $f_s/2$ Hz)
Needs: Filter BEFORE sampling.
Trap: Sample Q1: sin(Wt)/(πt) has cutoff W ⇒ largest W = π f_s.

$X_s(\omega) = (1/T_s)\Sigma\, X((\omega-2\pi\ell)/T_s)$; inside the band $X(\omega) = T_s X_s(T_s\,\omega)$
Trap: Mock Q8: peak 20 → 2, edge 2π/3 → 20π/3 rad/s.

Impulse train $p = \Sigma\delta(t-nT)$: $Z(\omega) = (1/T)\Sigma\, X(\omega-2\pi k/T)$
Needs: Copies every 2π/T rad/s.
Trap: Recover with a filter of gain T on the band (2024 Q6: H_2 = 1/4 for 5π<|ω|<8π).

Hz to rad/sample: $\omega = 2\pi f/f_s$
Needs: DT filter design.
Trap: 50 Hz at f_s=500 ⇒ π/5. Sketch DT filters over −2π..2π (they repeat).

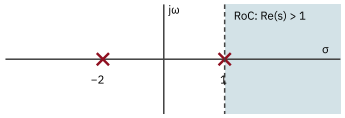
Energy of x_s: $\Sigma|x_s|^2 = (1/2\pi)\int_{-\pi}^{\pi}|X_s|^2d\omega$
Needs: Use evenness: (1/π)∫₀^π.
Trap: Mock Q8: 800/9.

Ideal reconstruction: H = T_s on |ω|<π/T_s; h = sinc, non-causal. ZOH delay ≈ $T_s/2$
Needs: Interpolation needs $h(nT_s) = \delta[n]$.

Write the reason: "$f_s > 2f_max$ so the DTFT copies do not overlap"
Trap: Marks are for this sentence.

10 Laplace: H(s), RoC, stability, output

2022 Q7, 2024 Q8, mock Q5, Q7, sample Q3, Q5



Causal: right of the rightmost pole. Stable: jω axis inside RoC.

ODE → Ĥ: d/dt → s, initial rest. y″+ay′+by = x ⇒ Ĥ = 1/(s²+as+b)
Trap: Poles −1, −2 ⇒ a=3, b=2.

Block diagram: name every wire in s, write one equation per adder, solve for Ŷ/X̂
Needs: 1/s is an integrator.

Causal ⇒ RoC: Re(s) > (rightmost pole)
Needs: All subsystems causal.
Trap: RoC never contains a pole.

BIBO stable ⇔ RoC contains the jω axis. Causal + stable ⇔ all poles Re < 0
Trap: Poles on the jω axis (±2j) are NOT stable.

Signal	Laplace RoC	Z RoC	Example
Right-sided	Re(s) > rightmost pole	|z| > outermost pole	$e^{-at}u(t) \Leftrightarrow 1/(s+a)$
Left-sided	Re(s) < leftmost pole	|z| < innermost pole	$-e^{-at}u(-t) \Leftrightarrow 1/(s+a)$
Two-sided	vertical strip	annulus	empty if poles cross
Finite length	all s	all z (except 0 / ∞)	FIR

Ŷ = Ĥ X̂, then partial fractions, then table (right-sided for causal)
Needs: Rational.
Trap: Divide first if degree num ≥ degree den.

Simple pole A/(s−p): A = [(s−p)Ŷ] at s=p. Repeated: A/(s+1)² + B/(s+1); get B by plugging s=0
Trap: 1/(s+a)² ⇔ $te^{-at}u(t)$.

Step response = inverse of Ĥ(s)/s. Final value (stable) = Ĥ(0)
Trap: Bounded step response does NOT prove BIBO stability.

Real system: complex poles come in conjugate pairs. "Input 1 gives output 0" ⇒ zero at s = 0

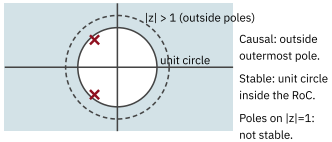
Past paper: 2024 Q8: Ĥ = (2s²−1)/((s−1)(s+2)), RoC Re(s)>1, not stable. x = $te^{-t}u+e^{-t}u$ ⇒ X̂=(s+2)/(s+1)² ⇒ y = −½$te^{-t}u$ + (7/4)$e^{-t}u$ + (1/4)$e^{t}u$.

Past paper: Sample Q5: Ĥ = 2s/(s²+4)+s/(s+1) ⇒ step response [sin 2t + e^{-t}]u(t); poles ±2j ⇒ not stable. Mock Q7: Ŷ=1/(s(s−1)) ⇒ y=(e^{t}−1)u, Ĥ= (s+2)/(s(s−1)).

Past paper: Sample Q3: RoC Re(s)>2, pole −1+j, real, 3 poles, input 1 → 0 ⇒ poles −1±j and 2, zero at 0, right-sided h.

11 Z-transform: H(z), poles, stability

2022 Q8, 2024 Q2–3, Q5, mock Q9, sample Q7



Difference eq. → Ĥ: x[n−m] → z^{-m}X̂, solve Ŷ/X̂
Trap: Then multiply top and bottom by z^2 to match a requested form.

Poles: roots of the denominator in z. Polar form $re^{j\theta}$
Trap: $z^2+z+1 = 0 \Rightarrow e^{\pm j2\pi/3}$ (on the unit circle).

Causal ⇒ |z| > outermost pole. Stable ⇔ unit circle in the RoC
Needs: Independent checks. FIR: poles only at 0 (always stable); IIR: any other pole.
Trap: Pole on |z|=1 ⇒ not stable. Step response: Ĥ(z)/(1−z^{-1}).

Inverse: match a table row, or partial fractions on Ŷ/z, then multiply by z
Trap: $\alpha^n\cos(\omega_o n)u[n]$ row: numerator $1-\alpha z^{-1}\cos\omega_o$.

Unstable ⇒ give a bounded input at the pole frequency (resonance)
Trap: x = cos(2πn/3) into poles $e^{\pm j2\pi/3}$ ⇒ y grows.

Ĥ → difference equation: cross-multiply, z^{-m} → delay m
Needs: Say "subject to initial rest".
Trap: Forgetting initial rest loses a mark.

x[n+1] in the equation ⇒ positive power of z in the numerator
Trap: 2022 Q8: Ĥ = z(2z+1)/(z+2), pole −2 ⇒ not stable.

Past paper: 2024 Q3: ½+z^{-1}−z^{-2} over 1+2z^{-1}+z^{-2} = (z^2+2z−2)/(2z^2+4z+2) ⇒ a=2, b=−2, c=2, d=4, e=2. Causal and LTI (initial rest).

Past paper: 2024 Q5: (1+0.5z^{-1})/(1+z^{-1}+z^{-2}): poles $e^{\pm j2\pi/3}$, RoC |z|>1, h = cos(2πn/3)u[n] (α=1, cos ω_o = −½), not stable.

Past paper: Mock Q9: $x=0.5^n u$, $y=0.25^n u \Rightarrow \hat{H}=(1-0.5z^{-1})/(1-0.25z^{-1})$, IIR, $h = 2\delta[n] - 0.25^n u[n]$. Sample Q7: step of $1/(1+z^{-1}+z^{-2})$ has Z-transform $1/(1-z^{-3})$, $|z|>1$.

12 Reference tables for any question

Standard results that work on any question, not only past papers.

System	h	H(ω)	H(s) or H(z)	Properties
Gain a	$a\delta$	a	a	memoryless, causal, stable
Delay τ (or N)	$\delta(t-\tau)$	$e^{-j\omega\tau}$	$e^{-s\tau}$ / z^{-N}	causal if $\tau \geq 0$, stable, all-pass
Integrator	$u(t)$	$1/(j\omega) + \pi\delta(\omega)$	$1/s$, $\mathrm{Re}(s) > 0$	causal, NOT stable
Differentiator	$\delta'(t)$	$j\omega$	s	NOT stable, high-pass
RC / 1st order CT	$(1/\tau)e^{-t/\tau}u(t)$	$1/(1+j\omega\tau)$	$1/(1+s\tau)$	low-pass, cutoff $1/\tau$, stable
Accumulator	$u[n]$	—	$1/(1-z^{-1})$, $\lvert z\rvert>1$	causal, NOT stable
First difference	$\delta[n]-\delta[n-1]$	$1-e^{-j\omega}$	$1-z^{-1}$	high-pass, FIR
M-point average	$(1/M)\Sigma_{k=0}^{M-1}$ $\delta[n-k]$	$e^{-j\omega(M-1)/2}$ $\sin(M\omega/2)/(M\sin(\omega/2))$	$(1/M)(1-z^{-M})/(1-z^{-1})$	low-pass, FIR, delay $(M-1)/2$
1st order DT	$a^n u[n]$	$1/(1-ae^{-j\omega})$	$1/(1-az^{-1})$, $\lvert z\rvert>\lvert a\rvert$	stable iff $\lvert a\rvert<1$; LPF if $a>0$, HPF if $a<0$
Ideal LPF	$\sin(Wt)/(\pi t)$	1 for $\lvert\omega\rvert<W$	—	non-causal

Operation	CTFT / DTFT	Laplace / Z	Fourier series
Shift t_0 / n_0	$e^{-j\omega t_0}X$ / $e^{-j\omega n_0}X$	$e^{-st_0}\hat{X}$ / $z^{-n_0}\hat{X}$	$e^{-jk\omega_0 t_0}X_k$
Multiply by $e^{j\omega_0 t}$	$X(\omega-\omega_0)$	$\hat{X}(s-j\omega_0)$ / $\hat{X}(e^{-j\omega_0}z)$	X_{k-M} ($\omega_0 = M\omega_fund$)
Convolution	XH	$\hat{X}\hat{H}$ (RoC $\supseteq \cap$)	$T_0 X_k Z_k$ (periodic conv.)
Product xz	$(1/2\pi)\, X * Z$	—	$\Sigma\, X_\ell Z_{k-\ell}$
d/dt or x[n]−x[n−1]	$j\omega X$ / $(1-e^{-j\omega})X$	$s\hat{X}$ / $(1-z^{-1})\hat{X}$	$jk\omega_0 X_k$ (CT only)
Time reversal	$X(-\omega)$	$\hat{X}(-s)$ / $\hat{X}(z^{-1})$	X_{-k}
Real signal	$X(-\omega) = X(\omega)^*$	poles/zeros conj. pairs	$X_{-k} = X_k^*$

Pole (CT / DT)	Time term	Behaviour
$s = -a$, $a>0$ / $z = r$, $0<r<1$	$e^{-at}u$ / $r^n u$	decays
$s = 0$ / $z = 1$	$u(t)$ / $u[n]$	constant, not stable
$\mathrm{Re}(s) > 0$ / $\lvert z\rvert > 1$	e^{at} / r^n, $r>1$	grows, not stable
$s = \pm j\omega_0$ / $z = e^{\pm j\theta}$	$\cos(\omega_0 t+\varphi)$ / $\cos(\theta n+\varphi)$	sustained oscillation, not stable
$s = -a \pm j\omega_0$ / $z = re^{\pm j\theta}$, $r<1$	$e^{-at}\cos(\omega_0 t+\varphi)$ / $r^n\cos(\theta n+\varphi)$	decaying oscillation
$z = -r$	$(-r)^n$	alternating sign (high frequency)
Repeated pole	extra factor t (CT) or n (DT)	te^{pt}, np^n

Partial-fraction case	Write	Find the coefficient
Distinct poles	$A/(s-p)$	$A = (s-p)F(s)$ at $s = p$ (cover-up)
Repeated $(s-p)^2$	$A/(s-p)^2 + B/(s-p)$	A by cover-up; B by plugging any s (e.g. 0)
Complex pair	$(Bs+C)/((s+b)^2+\omega_0^2)$	match $e^{-bt}\cos$ and $e^{-bt}\sin$ table rows
deg num ≥ deg den	divide first	quotient gives $\delta(t)$ (CT) or $\delta[n]$ (DT) terms
Z-transform	expand $\hat{X}(z)/z$	then multiply each term back by z

$\lvert H\rvert$ at ω = 0	$\lvert H\rvert$ at high ω (DT: ω = π)	Filter
large	small	low-pass
small	large	high-pass
small	small, peak between	band-pass
large	large, dip between	band-stop
equal everywhere		all-pass (pure delay)

DT check: $\omega = 0$ is $z = 1$ and $\omega = \pi$ is $z = -1$. CT check: $s = 0$ and $s \to \infty$

$e^{j\pi} = -1$, $e^{j\pi/2} = j$, $e^{j2\pi k} = 1$. $|a+jb| = \sqrt{(a^2+b^2)}$, $\angle(a+jb) = \mathrm{atan2}(b,a)$. $|1/(a+jb)| = 1/\sqrt{(a^2+b^2)}$

Trap: atan alone loses the quadrant when $a < 0$.

$\Sigma_{n=0}^{N-1} a^n = (1-a^N)/(1-a)$. $\Sigma_{n=0}^{\infty} a^n = 1/(1-a)$, $|a|<1$. $\Sigma_{n=0}^{\infty} n\, a^n = a/(1-a)^2$

cos/sin at 0, $\pi/6$, $\pi/4$, $\pi/3$, $\pi/2$: cos = 1, $\sqrt{3}/2$, $\sqrt{2}/2$, ½, 0; sin = 0, ½, $\sqrt{2}/2$, $\sqrt{3}/2$, 1

Useful formulas

Sifting formula:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \quad \text{for all } n \qquad\qquad x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau \quad \text{for all } t$$

Convolution:

$$(x_1 * x_2)[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k] \quad \text{for all } n \qquad\qquad (x_1 * x_2)(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)\,d\tau \quad \text{for all } t$$

Infinite sum: If $|\alpha| < 1$ then

$$\sum_{k=0}^{\infty} \alpha^k = \frac{1}{1-\alpha}.$$

Indefinite integrals:

$$\int e^x\,dx = e^x + c \qquad\qquad \int x^k\,dx = \frac{1}{k+1}x^{k+1} + c \quad (\text{for } k \neq -1)$$

DTFT of sampled signal If x has continuous-time Fourier transform X then $x_s[n] = x(nT)$ for all n has discrete-time Fourier transform

$$X_s(\omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X((\omega - 2\pi k)/T) \quad \text{for all } \omega.$$

Continuous-time Fourier series tables

If x is periodic with period $T_0 = 2\pi/\omega_0$ (and x is sufficiently regular) then

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t} \quad \text{where} \quad X_k = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-jk\omega_0 t}\,dt$$

Time domain signal $x(t)$ for $-T_0/2 \le t \le T_0/2$	Fourier series coefficients X_k of periodic extension $x_{\text{periodic}}(t)$	Comments
$e^{j\omega_0 t}$	$\delta[k-1]$	
$\cos(\omega_0 t)$	$\frac{1}{2}\left(\delta[k-1] + \delta[k+1]\right)$	
$\sin(\omega_0 t)$	$\frac{1}{2j}\left(\delta[k-1] - \delta[k+1]\right)$	
1	$\delta[k]$	
$[u(t+T) - u(t-T)]$	$\dfrac{\sin(k\omega_0 T)}{k\pi}$	Square wave
$\delta(t)$	$\dfrac{1}{T_0}$	Impulse train

Table 1: Table of Fourier series coefficients. In the table, all time domain signals are periodic with fundamental period T_0 and fundamental frequency $\omega_0 = 2\pi/T_0$ and the parameter T satisfies $0 < T < T_0/2$. The entry $u(t+T) - u(t-T)$ should be interpreted as a periodic square wave that agrees with $u(t+T) - u(t-T)$ for $-T_0/2 < t < T_0/2$. The entry δ should be interpreted as an impulse train with impulses at integer multiples of T_0.

Time domain $y(t) =$	Frequency-domain $Y_k =$	Name
$\alpha x(t) + \beta z(t)$	$\alpha X_k + \beta Z_k$	Linearity
$x(t-\tau)$	$e^{-jk\omega_0\tau} X_k$	Delay
$e^{jM\omega_0 t}x(t)$	X_{k-M}	Modulation
$\int_0^{T_0} x(\tau)z(t-\tau)\,d\tau$	$T_0 X_k Z_k$	Periodic convolution
$x(t)z(t)$	$\sum_{\ell=-\infty}^{\infty} X_\ell Z_{k-\ell}$	Multiplication
$x(t)^*$	X_{-k}^*	Conjugation
Time domain	**Frequency-domain**	**Name**
$x(t)$ is real	$X_k = X_{-k}^*$	Conjugate symmetry in frequency
$x(t) = x(-t)^*$	X_k is real	Conjugate symmetry in time
$\dfrac{1}{T_0}\int_0^{T_0} \lvert x(t)\rvert^2\,dt = \sum_{k=-\infty}^{\infty} \lvert X_k\rvert^2$		Parseval's relation

Table 2: Properties of continuous-time Fourier series. In the table, τ is a real number, M is an integer, and α and β are complex numbers. All time-domain signals are periodic with fundamental period T_0 and fundamental frequency $\omega_0 = 2\pi/T_0$. All integrals from 0 to T_0 can be replaced with integrals from a to $a + T_0$ for any real a.

Discrete-time Fourier series tables

If x is periodic with period $N_0 = 2\pi/\omega_0$ then

$$x[n] = \sum_{k=0}^{N_0-1} X_k e^{jk\omega_0 n} \quad \text{where} \quad X_k = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x[n] e^{-jk\omega_0 n}.$$

Time domain signal $x[n]$ for $-N_0/2 \le n < N_0/2$	Fourier series coefficients X_k of periodic extension $x_{\text{periodic}}[n]$	Comments
$e^{j\omega_0 n}$	$\delta[k-1]$	
$\cos(\omega_0 n)$	$\frac{1}{2}\left(\delta[k-1] + \delta[k+1]\right)$	
$\sin(\omega_0 n)$	$\frac{1}{2j}\left(\delta[k-1] - \delta[k+1]\right)$	
1	$\delta[k]$	
$u[n+N] - u[n-(N+1)]$	$\dfrac{\sin\left(k\omega_0(2N+1)/2\right)}{k\omega_0/2}$	Square wave
$\delta[n]$	$\dfrac{1}{N_0}$	Impulse train

Table 3: Table of discrete-time Fourier series coefficients. In the table, all time domain signals are periodic with fundamental period N_0 and fundamental frequency $\omega_0 = 2\pi/N_0$ and N is an integer with $0 < N < N_0/2$. All Fourier series coefficient sequences are also periodic with fundamental period N_0. The time-domain entry $u[n+N] - u[n-N]$ should be interpreted as an N_0-periodic square wave that agrees with $u[n+N] - u[n-N]$ for $-N_0/2 \le n < N_0/2$.

Time domain $y[n] =$	Frequency-domain $Y_k =$	Name
$\alpha x[n] + \beta z[n]$	$\alpha X_k + \beta Z_k$	Linearity
$x[n-N]$	$e^{-jk\omega_0 N} X_k$	Delay
$e^{jM\omega_0 n} x[n]$	X_{k-M}	Modulation
$\sum_{\ell=0}^{N_0-1} x[\ell] z[n-\ell]$	$N_0 X_k Z_k$	Periodic convolution
$x[n] z[n]$	$\sum_{\ell=0}^{N_0-1} X_\ell Z_{k-\ell}$	Multiplication
$x[n]^*$	X_{-k}^*	Conjugation
Time domain	Frequency-domain	Name
$x[n]$ is real	$X_k = X_{-k}^*$	Conjugate symmetry in frequency
$x[n] = x[-n]^*$	X_k is real	Conjugate symmetry in time
$\dfrac{1}{N_0} \sum_{n=0}^{N_0-1} \lvert x[n]\rvert^2 \, dt = \sum_{k=0}^{N_0-1} \lvert X_k\rvert^2$		Parseval's relation

Table 4: Properties of discrete-time Fourier series. In the table, N and M are integers, and α and β are complex numbers. Time-domain signals are periodic with fundamental period N_0 and fundamental frequency $\omega_0 = 2\pi/N_0$. Fourier series coefficients are periodic with fundamental period N_0. Sums from 0 to $N_0 - 1$ can be replaced with sums from a to $a + N_0 - 1$ for any integer a.

Continuous-time Fourier transform tables

If x is an absolutely summable (and sufficiently regular) continuous-time signal then

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,d\omega \quad \text{where} \quad X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

Time domain signal $x(t) =$	Continuous-time Fourier transform $X(\omega) =$
$e^{-at}u(t)$ (for $\mathrm{Re}(a) > 0$)	$\frac{1}{a+j\omega}$
$u(t+T_1) - u(t-T_1)$	$\frac{2\sin(\omega T_1)}{\omega}$
$\frac{\sin(Wt)}{\pi t}$	$u(\omega+W) - u(\omega-W)$
$\delta(t)$	1
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
$\delta(t-\tau)$	$e^{-j\omega\tau}$
$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$
$te^{-at}u(t)$ (for $\mathrm{Re}(a) > 0$)	$\frac{1}{(a+j\omega)^2}$

Table 5: Table of continuous-time Fourier transform pairs. In the table, the parameter W is a positive real number, the parameter a is a complex number, and the parameter τ is a real number.

Time domain $y(t) =$	Frequency-domain $Y(\omega) =$	Name
$\alpha x(t) + \beta z(t)$	$\alpha X(\omega) + \beta Z(\omega)$	Linearity
$x(t-\tau)$	$e^{-j\omega\tau}X(\omega)$	Delay
$e^{j\omega_0 t}x(t)$	$X(\omega-\omega_0)$	Modulation
$\int_{-\infty}^{\infty} x(\tau)z(t-\tau)\,d\tau$	$X(\omega)Z(\omega)$	Convolution
$x(t)z(t)$	$\frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega)Z(\omega-\Omega)\,d\Omega$	Multiplication
$x(t)^*$	$X(-\omega)^*$	Conjugation
$x(at)$	$\frac{1}{\lvert a\rvert}X(\omega/a)$	Time-scaling
Time domain	Frequency-domain	Name
$x(t)$ is real	$X(\omega) = X(-\omega)^*$	Conjugate symmetry in frequency
$x(t) = x(-t)^*$	$X(\omega)$ is real	Conjugate symmetry in time
$\int_{-\infty}^{\infty} \lvert x(t)\rvert^2\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} \lvert X(\omega)\rvert^2\,d\omega$		Parseval's relation

Table 6: Properties of continuous-time Fourier transform. In the table, τ is a real number, ω_0 is a real number, a is a non-zero real number, and α and β are complex numbers.

Discrete-time Fourier transform tables

If x is an absolutely summable discrete-time signal then

$$x[n] = \frac{1}{2\pi}\int_0^{2\pi} X(\omega)e^{j\omega n}\,d\omega \quad \text{where} \quad X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Time domain signal $x[n] =$	Discrete-time Fourier transform $X(\omega) =$
$\alpha^n u[n]$ (for $\lvert\alpha\rvert < 1$)	$\dfrac{1}{1-\alpha e^{-j\omega}}$
$u[n+N] - u[n-N]$	$\dfrac{\sin(N\omega)}{\sin(\omega/2)}e^{j\omega/2}$
$\dfrac{\sin(Wn)}{\pi n}$	2π-periodic extension of $u(\omega+W) - u(\omega-W)$
$\delta[n]$	1
$\delta[n-n_0]$	$e^{-j\omega n_0}$
$(n+1)\alpha^n u[n]$ (for $\lvert\alpha\rvert < 1$)	$\dfrac{1}{(1-\alpha e^{-j\omega})^2}$

Table 7: Table of discrete-time Fourier transform pairs. In the table, the parameter W satisfies $0 < W < \pi$, the parameter α is a complex number, and the parameter n_0 is an integer.

Time domain $y[n] =$	Frequency-domain $Y(\omega) =$	Name
$\alpha x[n] + \beta z[n]$	$\alpha X(\omega) + \beta Z(\omega)$	Linearity
$x[n-N]$	$e^{-j\omega N}X(\omega)$	Delay
$e^{j\omega_0 n}x[n]$	$X(\omega-\omega_0)$	Modulation
$\sum_{\ell=-\infty}^{\infty} x[\ell]z[n-\ell]$	$X(\omega)Z(\omega)$	Convolution
$x[n]z[n]$	$\frac{1}{2\pi}\int_0^{2\pi} X(\Omega)Z(\omega-\Omega)\,d\Omega$	Multiplication
$x[n]^*$	$X(-\omega)^*$	Conjugation
Time domain	Frequency-domain	Name
$x[n]$ is real	$X(\omega) = X(-\omega)^*$	Conjugate symmetry in frequency
$x[n] = x[-n]^*$	$X(\omega)$ is real	Conjugate symmetry in time
$\displaystyle\sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2\,dt = \frac{1}{2\pi}\int_0^{2\pi} \lvert X(\omega)\rvert^2\,d\omega$		Parseval's relation

Table 8: Properties of discrete-time Fourier transform. In the table, N is an integer, ω_0 is a real number, and α and β are complex numbers. All discrete-time Fourier transforms are periodic with fundamental period 2π.

Laplace transform tables

If x is a (sufficiently regular) continuous-time signal then

$$\hat{X}(s) = \int_{-\infty}^{\infty} x(t) e^{-st}\, dt$$

Time domain signal $x(t) =$	Laplace transform $\hat{X}(s) =$	Region of convergence $\mathrm{RoC}(x)$
$\delta(t)$	1	$\mathbb{C}$
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -\mathrm{Re}(a)$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -\mathrm{Re}(a)$
$\delta(t-\tau)$	$e^{-s\tau}$	$\mathbb{C}$
$e^{-bt}\cos(\omega_0 t)u(t)$	$\frac{s+b}{(s+b)^2+\omega_0^2}$	$\mathrm{Re}(s) > -b$
$e^{-bt}\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{(s+b)^2+\omega_0^2}$	$\mathrm{Re}(s) > -b$

Table 9: Table of Laplace transform pairs. In the table, the parameter a is a complex number, and the parameters b and τ and ω_0 are real numbers.

Time domain $y(t)$	s-domain $\hat{Y}(s)$	Region of convergence $\mathrm{RoC}(y)$	Name
$\alpha x_1(t) + \beta x_2(t)$	$\alpha\hat{X}_1(s) + \beta\hat{X}_2(s)$	$\mathrm{RoC}(y) \supseteq \mathrm{RoC}(x_1) \cap \mathrm{RoC}(x_2)$	Linearity
$x(t-\tau)$	$e^{-s\tau}\hat{X}(s)$	$\mathrm{RoC}(y) = \mathrm{RoC}(x)$	Delay
$(x_1 * x_2)(t)$	$\hat{X}_1(s)\hat{X}_2(s)$	$\mathrm{RoC}(y) \supseteq \mathrm{RoC}(x_1) \cap \mathrm{RoC}(x_2)$	Convolution
$x(t)^*$	$[\hat{X}(s^*)]^*$	$\mathrm{RoC}(y) = \mathrm{RoC}(x)$	Conjugation
$x(at)$	$\frac{1}{\lvert a\rvert}\hat{X}\left(\frac{s}{a}\right)$	$\mathrm{RoC}(y) = \{s \in \mathbb{C} : s/a \in \mathrm{RoC}(x)\}$	Time scaling
$tx(t)$	$-\frac{d}{ds}\hat{X}(s)$	$\mathrm{RoC}(y) = \mathrm{RoC}(x)$	Multiplication by t
$e^{s_0 t}x(t)$	$\hat{X}(s-s_0)$	$\mathrm{RoC}(y) = \{s \in \mathbb{C} : s - s_0 \in \mathrm{RoC}(x)\}$	Mult. by exponential
$\int_{-\infty}^{t} x(\tau)\, d\tau$	$\hat{X}(s)/s$	$\mathrm{RoC}(y) \supseteq \mathrm{RoC}(x) \cap \{s \in \mathbb{C} : \mathrm{Re}(s) > 0\}$	Integration
$\frac{d}{dt}x(t)$	$s\hat{X}(s)$	$\mathrm{RoC}(y) \supseteq \mathrm{RoC}(x)$	Differentiation

Table 10: Properties of the Laplace transform. In the table, τ is a real number, a is a non-zero real number, and s_0 and α and β are complex numbers.

Z-transform tables

If x is a discrete-time signal then

$$\hat{X}(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Time domain signal $x[n] =$	Z-transform $\hat{X}(z) =$	Region of convergence RoC(x)
$\delta[n]$	1	$\mathbb{C}$
$u[n]$	$\frac{1}{1-z^{-1}}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert > 1\}$
$nu[n]$	$\frac{z^{-1}}{(1-z^{-1})^2}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert > 1\}$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert > \lvert a\rvert\}$
$a^n u[-n]$	$\frac{1}{1-a^{-1}z}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert < \lvert a\rvert\}$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert > \lvert a\rvert\}$
$\delta[n-m]$	z^{-m}	$\lvert z\rvert > 0$
$a^n \cos[\omega_0 n]u[n]$	$\frac{1-az^{-1}\cos(\omega_0)}{1-2az^{-1}\cos(\omega_0)+a^2z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$
$a^n \sin[\omega_0 n]u[n]$	$\frac{az^{-1}\sin(\omega_0)}{1-2az^{-1}\cos(\omega_0)+a^2z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$

Table 11: Table of Z-transform pairs. In the table, the parameter a is a complex number, and the parameter ω_0 is a real number.

Time domain $y[n] =$	Z-transform $\hat{Y}(z) =$	Region of convergence RoC(y)	Name
$\alpha x[n] + \beta z[n]$	$\alpha\hat{X}(z) + \beta\hat{Z}(z)$	RoC(y) $\supseteq$ RoC(x_1) $\cap$ RoC(x_2)	Linearity
$x[n-m]$	$z^{-m}\hat{X}(z)$	RoC(x) $\setminus \{0\}$	Delay
$a^n x[n]$	$\hat{X}(z/a)$	aRoC(x)	Modulation
$\sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]$	$\hat{X}_1(z)\hat{X}_2(z)$	RoC(y) $\supseteq$ RoC(x_1) $\cap$ RoC(x_2)	Convolution
$x[n]^*$	$\hat{X}(z^*)^*$	RoC(y) = RoC(x)	Conjugation
$x[-n]$	$\hat{X}(z^{-1})$	RoC(y) = $\{z \in \mathbb{C} : z^{-1} \in \text{RoC}(x)\}$	Time-reversal

Table 12: Properties of the Z-transform. In the table, m is an integer, a, α and β are complex numbers.